

Fluid Mechanics

Building Models to Describe our World

Pressure, Hydrostatics/Dynamics and Bernoulli's Principle



Outline

Pressure

- Pressure in a fluid
- Equilibireum
- Pressure gradient
- Atmospheric pressure
- Pascal's principle

Hydrostatics/Dynamics

- Buoyancy
- Origin of buoyancy force
- Hydrodynamics - flow rate
- Continuity

Bernoulli's Principle

- Definition and derivation
- Example: strong winds
- Viscosity
- Poiseuille flow
- Calculating water tower height
results

Pressure

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Pressure

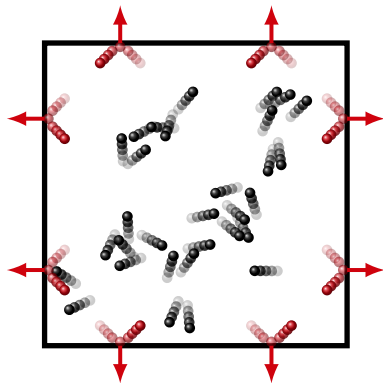
- Defined as force per unit area:

$$P = \frac{F}{A}$$

- Scalar quantity.
- Exerts a force that is normal to any surface.

Pressure in a fluid

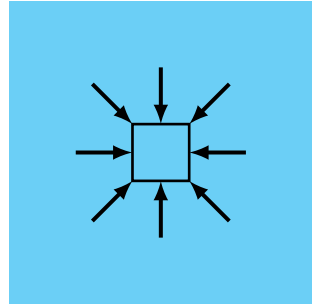
- Pressure is a scalar, it does not have a direction.
- A fluid (liquid or gas) cannot transmit force in a particular direction.
- The pressure in a fluid is in all directions.
- If there is a boundary on the fluid (e.g. a container), then the force from the pressure in the fluid is normal to the surface
- You can think of the particles in the fluid as bouncing against the surfaces. The force required to change their momentum is the reaction force to the force associated with their pressure.



Particles in the fluid bounce against the boundaries exerting an outward normal force.

Hydrostatic equilibrium

- Literally: Water not move equilibrium.
- Think of an element of fluid that is not moving.



A fluid element in hydrostatic equilibrium.

Pressure gradient in a fluid

- In order for the fluid element to be in equilibrium, the net force on it must be zero:

$$\sum F_y = P_2 A - mg - P_1 A = 0$$

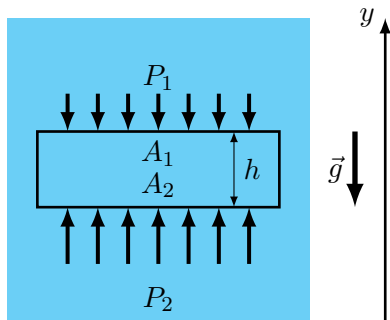
- Using density for the mass:

$$m = \rho V = \rho Ah$$

- Altogether:

$$P_2 A - \rho Ahg - P_1 A = 0$$

$$P_2 - P_1 = \rho gh$$



A fluid element in hydrostatic equilibrium, with gravity.

Pressure gradient in a fluid

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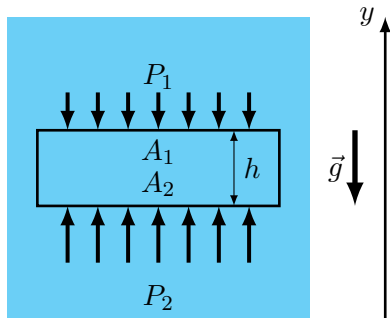
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A fluid element in hydrostatic equilibrium, with gravity.

In other words, the pressure increases as you move down:

$$P_2 = P_1 + \rho gh$$

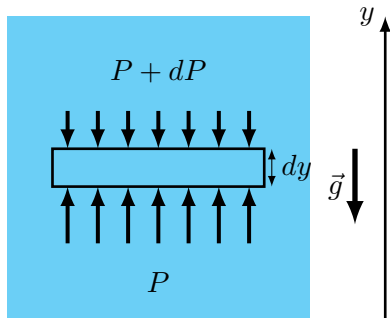
If the fluid is *compressible*

- The density of the fluid can vary with pressure (e.g. if the fluid is a gas, like the Earth's atmosphere).
- Then, we model an infinitesimal fluid element, small enough so that density is constant:

$$\frac{dP}{dy} = -\rho g$$
$$\therefore P(y_2) - P(y_1) = \int_{y_1}^{y_2} \rho(y) g dy$$

- For constant density, recover:

$$\int_{y_1}^{y_2} \rho(y) g dy = \rho g (y_2 - y_1) = \rho g h$$



A compressible fluid element in hydrostatic equilibrium. Note that dP is negative in the figure.

The Earth's atmosphere

- A rough approximation is that density is proportional to pressure (e.g. ideal gas law).

$$\rho(y) = aP(y)$$

where a is just a constant of proportionality.

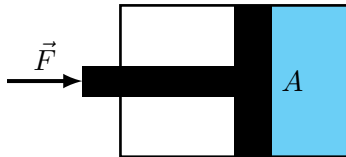
- One finds that:

$$\begin{aligned}\frac{dP}{dy} &= -\rho g = -aP(y)g \\ \therefore P(y) &= P_0 e^{-agy}\end{aligned}$$

- Pressure decreases exponentially with increasing altitude.

Pascal's principle

- If you exert an external pressure on a fluid, the pressure in the fluid increases everywhere by the same amount.



Pascal's principle: applying pressure externally to a fluid increases its pressure.

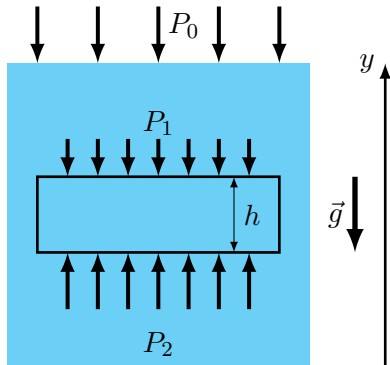
Pascal's principle

- If you exert an external pressure on a fluid, the pressure in the fluid increases everywhere by the same amount.
- The absolute pressure in, say, water, then depends on the atmospheric pressure, P_0 :

$$P_2 = P_0 + P_1 + \rho gh$$

- Or, in general, some distance h below the surface:

$$P_2 = P_0 + \rho gh$$



A fluid with atmospheric pressure on it.

Hydrostatics and Dynamics

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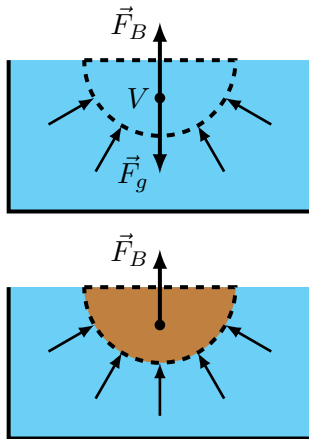
Pressure, Hydrostatics/Dynamics and Bernoulli's Principle

Buoyancy

- Same effect as hydrostatic equilibrium.
- There is a net upwards force on the fluid element because of the pressure gradient (due to gravity – pressure increases downwards).
- If you replace the fluid element with an object, that pressure gradient and resulting upwards force does not go away!
- Magnitude of the force is equal to the weight of the fluid displaced:

$$F_B = m_{fluid}g = \rho V g$$

- This does not change with depth!



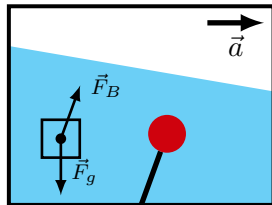
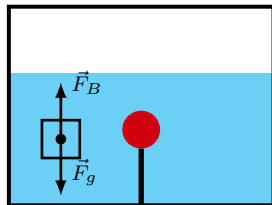
A volume, V , supported by the same buoyancy force, regardless of being filled by fluid or an object.

Buoyancy force comes from pressure gradient

- At rest, there is a pressure gradient in the air that is downwards because of gravity.
- When the car is accelerating, each element of the air must also accelerate with the car:
 - The sum of the forces must be:

$$\sum F = ma = \rho V a$$

- The only way this can happen is if the pressure gradient is in a direction to accelerate the fluid with the car (must oppose gravity and go to the right).

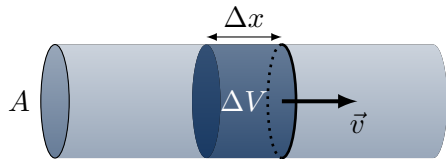


A fluid accelerating in a container with a floating balloon. FBD of forces on balloon.

Hydrodynamics - Flow rate, Q

- Think of liquid moving through a horizontal pipe of cross section, A .
- Continuity:
 - What comes in must come out!
 - If we put in 10 l/min into a pipe, then 10 l/min must come out the other end (incompressible)
- If the liquid flows at a speed, v , it covers a distance, Δx in a time, Δt .
- The volume of fluid flowing per unit time is called the flow rate:

$$Q = \frac{\Delta V}{\Delta t} = \frac{\Delta x A}{\Delta t} = Av$$



A fluid element, of volume, Δv , moving in a pipe.

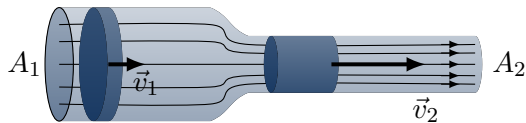
Continuity

- What goes in must come out!
- The flow rate in any point of the pipe must be the same:

$$Q_1 = Q_2$$

$$A_1 v_1 = A_2 v_2$$

$$\therefore v_2 = \frac{A_1}{A_2} v_1$$



A fluid element moving in a pipe that gets narrower, laminar flow.

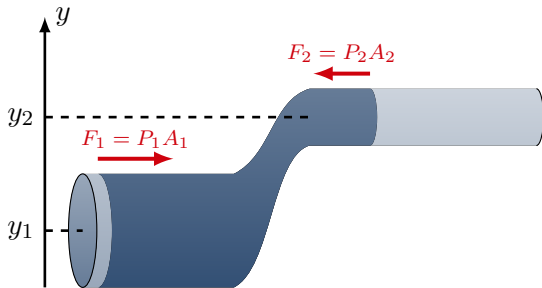
Bernoulli's Principle

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Bernoulli's Principle

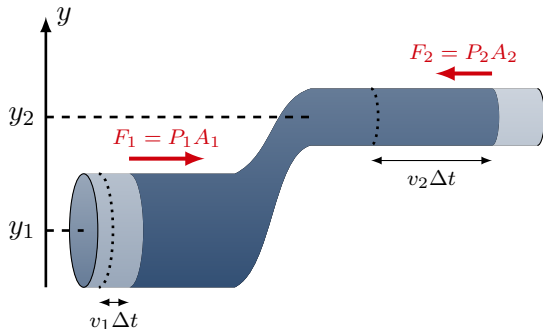
- Describes the laminar flow of an incompressible fluid with no friction (no viscosity).
- Model the motion of a fluid element as it moves through a pipe that changes height and cross-section (most general case).



A fluid element moving in a pipe that gets narrower while changing height. The motion is driven by a change in pressure (from high to low).

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A fluid element moving in a pipe that gets narrower while changing height. The motion is driven by a change in pressure (from high to low). After a time Δt

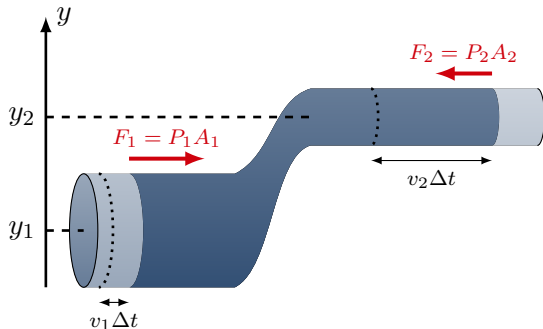
Bernoulli's Principle

- Apply Work-Energy Theorem to the fluid element.
- The force from the pressure at the bottom does positive work on the fluid element:

$$W_1 = F_1 \Delta x = (P_1 A_1)(v_1 \Delta t)$$

- The force from the pressure at the top does negative work on the fluid element:

$$W_2 = -F_2 \Delta x = -(P_2 A_2)(v_2 \Delta t)$$



A fluid element moving in a pipe that gets narrower while changing height. The motion is driven by a change in pressure (from high to low).

Bernoulli's Principle

- Gravity does negative work:

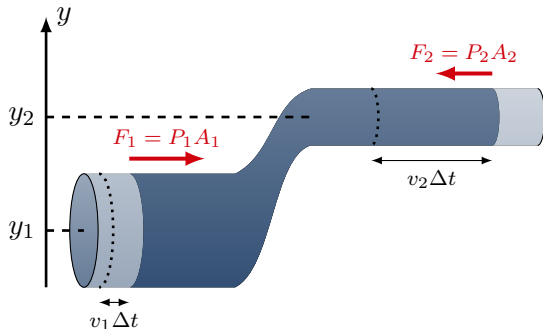
$$W_g = -\Delta m g (y_2 - y_1)$$

where the fluid element has mass:

$$\Delta m = \rho V = \rho A_1 v_1 \Delta t = \rho A_2 v_2 \Delta t$$

- The total work done on the fluid element must equal the change in kinetic energy:

$$W_1 + W_2 + W_g = \Delta K = \frac{1}{2} \Delta m (v_2^2 - v_1^2)$$



A fluid element moving in a pipe that gets narrower while changing height. The motion is driven by a change in pressure (from high to low).

Bernoulli's Principle

- All together (note $A_1 v_1 \Delta t = A_2 v_2 \Delta t$):

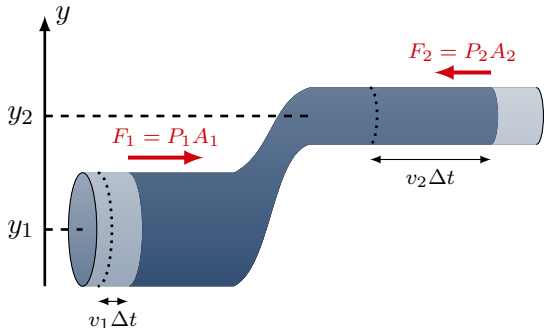
$$(P_1 A_1)(v_1 \Delta t) - (P_2 A_2)(v_2 \Delta t) - \rho A_1 v_1 \Delta t g(y_2 - y_1) = \frac{1}{2} \rho A_1 v_1 \Delta t (v_2^2 - v_1^2)$$

$$\therefore P_1 + \frac{1}{2} \rho v_1^2 + \rho g y_1 = P_2 + \frac{1}{2} \rho v_2^2 + \rho g y_2$$

- This looks a lot like conservation of mechanical energy (*it is!*):

$$E = K_1 + U_1 = K_2 + U_2 = \text{const.}$$

$$E = P + \frac{1}{2} \rho v^2 + \rho g y = \text{const.}$$

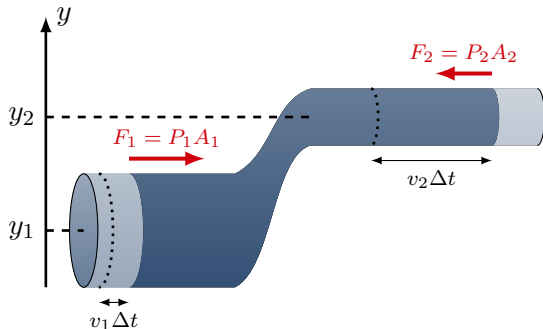


A fluid element moving in a pipe that gets narrower while changing height. The motion is driven by a change in pressure (from high to low).

Bernoulli's Principle

$$E = P + \frac{1}{2}\rho v^2 + \rho gy = \text{const.}$$

- Think of $\frac{1}{2}\rho v^2$ as the kinetic energy of the fluid (energy per volume).
- Think of ρgy as grav potential energy (per volume).
- Think of P as an additional potential energy (it's really the work being done by pressure, per unit volume).



A fluid element moving in a pipe that gets narrower while changing height. The motion is driven by a change in pressure (from high to low).

Recall: Flow rate and continuity

- Define (volume) flow rate, Q :

$$Q = \frac{\Delta V}{\Delta t} = \frac{\Delta x A}{\Delta t} = Av$$

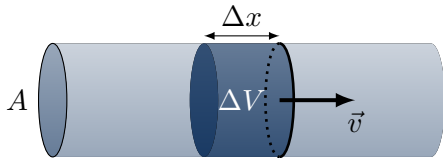
- Continuity:
 - What comes in must come out (incompressible):

$$\therefore A_1 v_1 = A_2 v_2$$

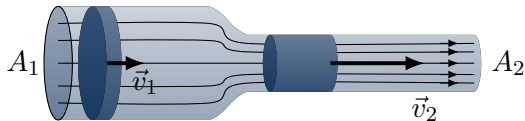
- For *compressible* fluid, use the mass instead of the volume:

$$\frac{\Delta m}{\Delta t} = \rho \frac{\Delta V}{\Delta t} = \rho Av$$

$$\therefore \rho_1 A_1 v_1 = \rho_2 A_2 v_2$$



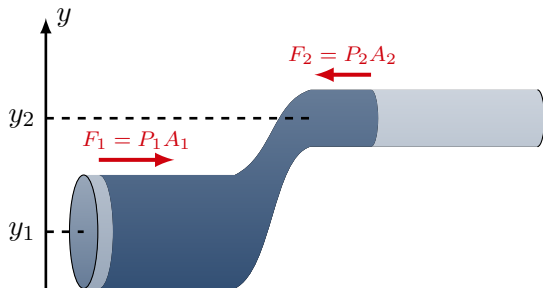
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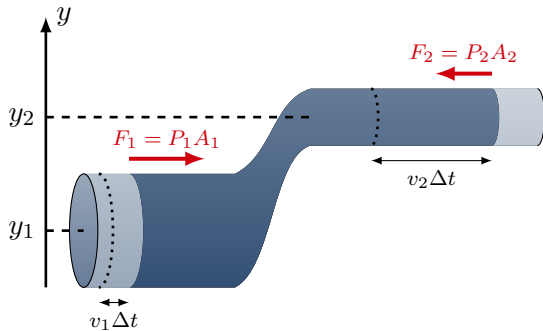


A fluid element moving in a pipe that gets narrower while changing height.

Bernoulli's Principle

$$E = P + \frac{1}{2}\rho v^2 + \rho gy = \text{const.}$$

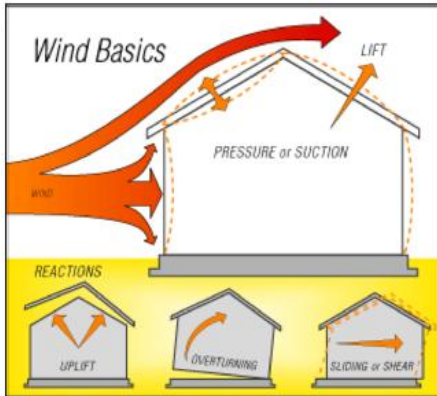
- Think of $\frac{1}{2}\rho v^2$ as the kinetic energy of the fluid (energy per volume).
- Think of ρgy as gravitational potential energy (per volume).
- Think of P as an additional potential energy (per unit volume) due to the pressure in the fluid. It wants to flow where that energy (pressure) is lower.



A fluid element moving in a pipe that gets narrower while changing height. After a time Δt

Example: Strong Winds

- This is how strong winds blow roods off your house!

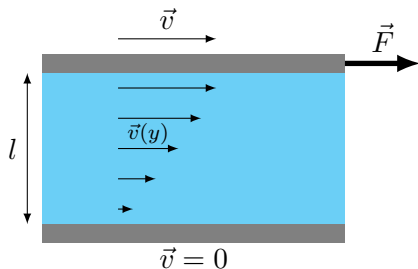


Viscosity

- In reality, fluids have “viscosity”. Think of maple syrup as being more viscous than water.
- Viscosity is the results of internal friction of the fluid molecules with each other.
- We can define it based on imagining a stationary fluid between 2 plates and measuring the force required to pull the top plate at constant speed v :

$$F \propto \frac{Av}{l} \rightarrow F = \eta \frac{Av}{l}$$

- A pressure difference is required to push fluid through a pipe, since the fluid “sticks” to the walls. Think: “syringe full of maple syrup”.



Fluid between two plates. The bottom plate is stationary and the one on top is pulled with a force, $\vec{F}$, at a constant speed, v . Different layers in the fluid move at different speeds.

Poiseuille flow

- The flow rate is proportional to the pressure difference across the pipe:

$$Q \propto \Delta P$$

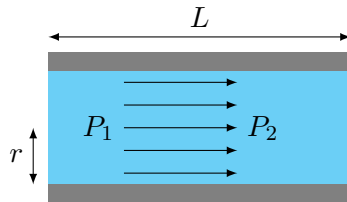
- The constant of proportionality is called the resistance of the pipe (one over R , actually):

$$Q = \frac{1}{R} \Delta P$$

- We can write this as:

$$\Delta P = QR$$

- For a horizontal pipe: $R = \frac{8\eta L}{\pi r^4}$

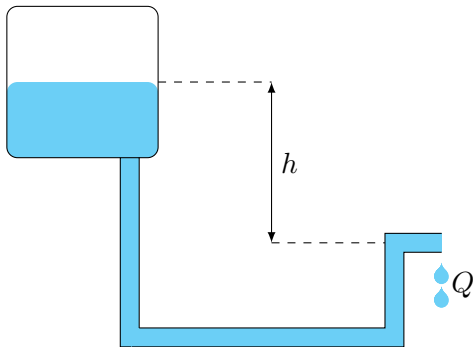


Poiseuille flow in a pipe due to a pressure difference.

Consequences

- If there is no flow through the pipe, then there is no drop in pressure along the pipe.
- If there is flow through the pipe, the pressure must drop along the pipe.

A model for your experiment



Water from a tower at a height, h , above your faucet results in a flow rate Q out of the faucet. By measuring, Q , one should be able to determine, h ...

Step 1: Measure Q, determine speed of water

- In my bathroom, I filled a 500 ml cup in 7 s:

$$500 \text{ ml} = 0.0005 \text{ m}^3$$

$$\therefore Q = \frac{0.0005 \text{ m}^3}{7 \text{ s}} = 7.14 \times 10^{-5} \text{ m}^3/\text{s}$$

- Radius of faucet is about 1 cm:

$$\therefore A = \pi r^2 = 3.1 \times 10^{-4} \text{ m}^2$$

- Determine speed of water coming out of faucet:

$$v = \frac{Q}{A} = 0.23 \text{ m/s}$$

Step 2: Apply Bernoulli

- Need to apply:

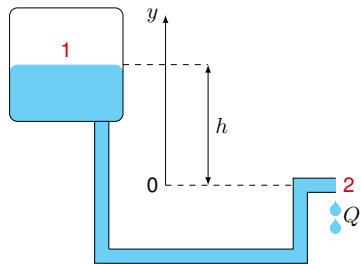
$$P_1 + \frac{1}{2}\rho_1 v_1^2 + \rho g y_1 = P_2 + \frac{1}{2}\rho_2 v_2^2 + \rho g y_2$$

- Choose points 1 and 2:

- $P_1 = P_2 = P^{atm}$
- $y_1 = h, y_2 = 0$
- $v_1 = 0, v_2 = 0.23 \text{ m/s}$

$$\therefore \rho g h = \frac{1}{2} \rho v_2^2$$

$$\therefore h = \frac{v_2^2}{2g} = 2.6 \text{ mm} \quad !!!$$



Choosing two locations to apply Bernoulli's principle.

Step 3: Does this make sense?

- Clearly, it does not, the water tower is more than 2.6 m above my faucet! There's a much greater height between my bathroom and my basement, and the water doesn't explode out of my basement faucet!
- However, it does make sense that the water wouldn't have to fall from very high in order to reach a speed of $v = 0.23 \text{ m/s}$...
- When water is not flowing, the pressure from the water height of 2.6 m at the faucet would be:

$$P = \rho gh = 25.8 \text{ Pa}$$

- This means you could plug the faucet by exerting a force F with your finger:

$$F = PA = 0.0008 \text{ N}$$

- That's a small force, you should easily be able to plug the faucet with your thumb!
- Give it a try. Not so easy, so clearly the calculation above is not a model of the real world... so what gives?

Your faucet in the real world...

- Water does not “just” need to flow in the pipes due to gravity.
- It must also overcome a lot of friction in the pipes, due to its viscosity. Bernoulli’s principle completely ignores this.
- You need a large pressure difference (50 psi = 344,738 Pa is typical) to actually drive the water through the pipes.
- Something like Poiseuille flow, but more complicated, since not just a simple horizontal pipe.

$$\Delta P = QR$$

- Need a large ΔP to get a large Q if R is large.
- Remember, when there is no water flowing, $\Delta P = 0$, so when you try to plug the faucet with your thumb, you are feeling the P that comes from the “true” ρgh that is actually driving the water.
- Bernoulli will only make sense when the resistance of the pipe (or viscosity of the fluid) is negligible. E.g a large diameter, short pipe.